

# 1 Parameterized Quarto Reports

## 1.1 Learning objectives

After completing this chapter, you will be able to:

- Define parameters in a Quarto document's YAML header
- Access parameters in R code chunks to drive data retrieval and analysis
- Render parameterized reports from the command line and from scripts
- Build a template for journal-level bibliometric profiles
- Generate multiple reports in batch using `purrr` or shell loops

## 1.2 Setup

```
library(tidyverse)
library(openalexR)
library(glue)
library(gt)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

## 1.3 Conceptual background

Many bibliometric tasks are repetitive: generating the same analysis for 10 journals, 5 departments, or 20 funders. Writing separate scripts for each entity is error-prone and hard to maintain. **Parameterized reports** solve this by defining variables (parameters) that change the report's content without changing its structure.

Quarto supports parameters via the `params` field in the YAML header. Parameters can be strings, numbers, or lists. Inside the document, `params$journal_id` (or `params$institution`) drives data fetching, filtering, and labelling. At render time, parameter values are passed via the command line or a rendering script.

This approach scales naturally: a loop or `purrr::walk()` call renders dozens of reports with different parameters, producing a complete portfolio of bibliometric profiles with consistent methodology and formatting.

Parameterized reports are especially valuable for institutional reporting (annual bibliometric profiles for each department), journal analytics (quarterly reports for a publisher's journal portfolio), and funding assessment (per-grant output summaries).

## 1.4 Worked example

### 1.4.1 Defining parameters

A parameterized Quarto document starts with parameters in the YAML header:

```
# Example YAML header (not executable R):
#
# ---
# title: "Journal Bibliometric Profile"
# params:
#   journal_id: "S148561398"
#   journal_name: "Scientometrics"
#   start_year: 2020
#   end_year: 2023
# ---
```

### 1.4.2 Using parameters in code

```
journal_id <- "S148561398"
journal_name <- "Scientometrics"
start_year <- 2020
end_year <- 2023
```

```
works <- oa_fetch(
  entity = "works",
  primary_location.source.id = journal_id,
  from_publication_date = paste0(start_year, "-01-01"),
  to_publication_date = paste0(end_year, "-12-31"),
  type = "article",
  options = list(sample = 300, seed = 42)
)

cat(glue("Journal: {journal_name}\n"))
```

```
#> Journal: Scientometrics
```

```
cat(glue("Period: {start_year}--{end_year}\n"))
```

```
#> Period: 2020--2023
```

```
cat(glue("Articles retrieved: {nrow(works)}\n"))
```

```
#> Articles retrieved: 300
```

### 1.4.3 Automated summary statistics

```
summary_stats <- works |>
  summarise(
    total_articles = n(),
    mean_citations = round(mean(cited_by_count), 1),
    median_citations = median(cited_by_count),
    h_index = compute_h_index(cited_by_count),
```

| Metric           | Value |
|------------------|-------|
| total_articles   | 300   |
| mean_citations   | 19.4  |
| median_citations | 12    |
| h_index          | 35    |
| oa_rate          | 46.3  |

```

    oa_rate = round(mean(oa_status != "closed", na.rm = TRUE) * 100, 1)
  ) |>
  mutate(across(everything(), as.character)) |>
  pivot_longer(everything(), names_to = "Metric", values_to = "Value")

summary_stats |> gt()

```

#### 1.4.4 Parameterized figure

```

works |>
  mutate(year = year(publication_date)) |>
  count(year) |>
  ggplot(aes(x = factor(year), y = n)) +
  geom_col(fill = palette_sci(1)) +
  labs(x = "Year", y = "Articles",
       title = glue("{journal_name} - Annual Output")) +
  theme_sci()

```

#### 1.4.5 Batch rendering

```

# Render for multiple journals:
journals <- tibble(
  journal_id = c("S148561398", "S202381698", "S137773608"),
  journal_name = c("Scientometrics", "PLOS ONE", "Nature")
)

walk2(journals$journal_id, journals$journal_name, function(jid, jname) {
  quarto::quarto_render(
    input = "journal_profile.qmd",
    execute_params = list(
      journal_id = jid,
      journal_name = jname,
      start_year = 2020,
      end_year = 2023
    ),
    output_file = glue("reports/{jname}_profile.html")
  )
})

```

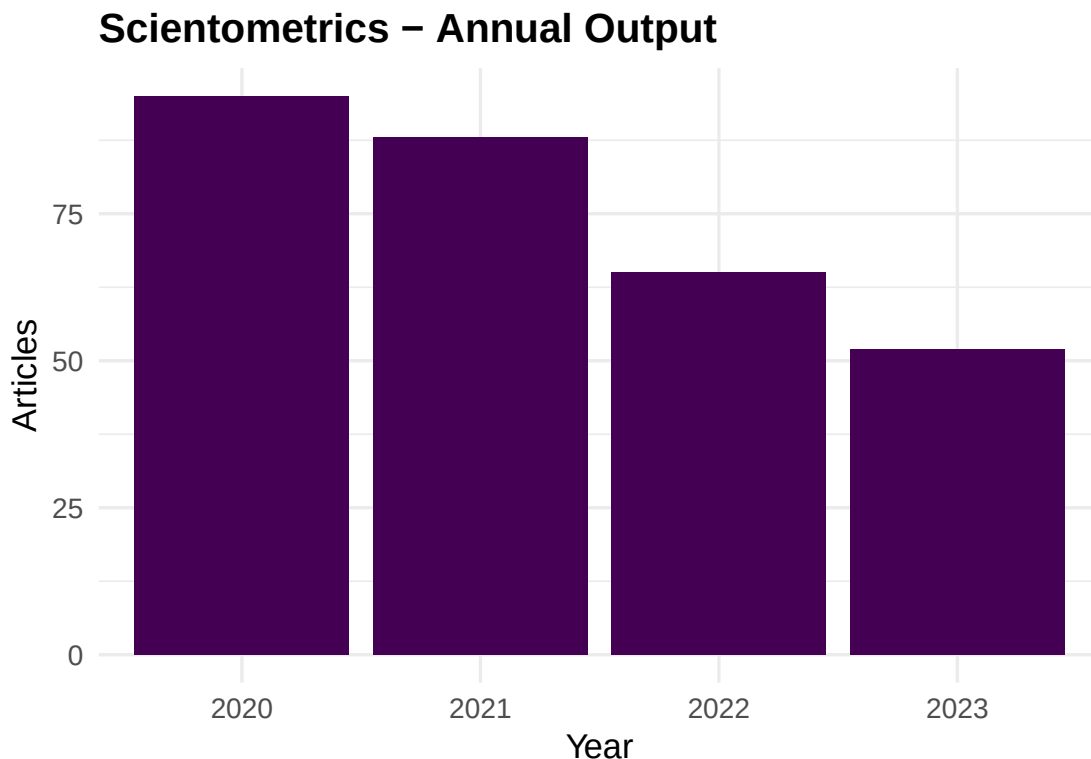


Figure 1: Annual publication count for the selected journal.

## 1.5 Diagnostics and interpretation

- **Parameter validation:** Check that parameters produce valid API queries. Invalid journal IDs return empty results silently.
- **Sample consistency:** When using `options = list(sample = N)`, the sample changes if the underlying corpus changes. Use `seed` for stability.
- **Output naming:** Automated file naming should avoid special characters. Sanitise journal names before using them in file paths.
- **Error handling:** Batch rendering should catch and log errors for individual reports without stopping the entire batch.

## 1.6 Limitations and responsible use

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- **Automated mindless.** Parameterized reports produce consistent output but still require human review. An automated report with a data error will produce the wrong results consistently across all entities.
- **One template fits all?** Different entities may need different analyses. A template designed for a large journal may produce misleading results for a small one (e.g., sparse yearly bins).
- **API costs.** Batch rendering for 100 journals makes 100 API calls. Respect rate limits and cache results.
- **Report fatigue.** Generating too many reports can overwhelm decision-makers. Produce reports for entities that need them, not for every possible parameter value [hicks2015].

## 1.8 Common pitfalls

## 1.9 Common pitfalls

- **Hardcoding values that should be parameters.** If you find yourself editing a date or ID in the script, it should be a parameter.
- **Not testing with extreme parameters.** A journal with 5 papers per year needs different visualisations than one with 5,000. Test edge cases.
- **Rendering without caching.** Each render re-executes all code. Use Quarto's `cache: true` or file-based caching to avoid redundant API calls.
- **Forgetting output format compatibility.** Interactive widgets work in HTML but not PDF. Ensure your template degrades gracefully.

## 1.10 Exercises

1. **Institutional template.** Create a parameterized Quarto document for institutional profiles. Parameters: institution ID, name, start year, end year. Include publication count, citation distribution, and top authors.
2. **Batch rendering.** Render the journal profile template for 3 different journals. Compare the resulting reports side by side.
3. **Conditional content.** Add conditional logic: if the sample has fewer than 50 papers, display a warning message instead of the citation distribution figure.

## 1.11 Solutions

Solutions are provided in 1.11.

## 1.12 Further reading

- @aria2017 — `bibliometrix` reporting features.
- @priem2022 — OpenAlex API for programmatic data retrieval in reports.

## 1.13 Session info

```
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#>
#> Matrix products: default
#> BLAS: /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblas-p-r0.3.26.so; LAPACK version 3.12.0
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8      LC_NUMERIC=C          LC_TIME=C.UTF-8      LC_COLLATE=C.UTF-8   LC_
#>  [6] LC_MESSAGES=C.UTF-8  LC_PAPER=C.UTF-8     LC_NAME=C            LC_ADDRESS=C         LC_
#> [11] LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
```

```

#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats      graphics  grDevices  utils      datasets  methods   base
#>
#> other attached packages:
#> [1] DT_0.34.0          plotly_4.12.0      uwot_0.2.4         Matrix_1.7-0
#> [5] word2vec_0.4.1     stm_1.3.8          topicmodels_0.2-17 quanteda.textstat
#> [9] visNetwork_2.1.4   gggraph_2.2.2      tidygraph_1.3.1    igraph_2.3.1
#> [13] quanteda_4.4        pdftools_3.9.0     arrow_24.0.0       bibliometrix_5.4.0
#> [17] RefManager_1.4.0   bib2df_1.1.2.0     rcrossref_1.2.1    gt_1.3.0
#> [21] tidytext_0.4.3     glue_1.8.1         openalexR_3.0.1    lubridate_1.9.5
#> [25] forcats_1.0.1      stringr_1.6.0      dplyr_1.2.1        purrr_1.2.2
#> [29] readr_2.2.0        tidyr_1.3.2        tibble_3.3.1       ggplot2_4.0.3
#> [33] tidyverse_2.0.0    bookdown_0.46      fs_2.1.0           rmarkdown_2.31
#>
#> loaded via a namespace (and not attached):
#> [1] bibtex_0.5.2       RColorBrewer_1.1-3 jsonlite_2.0.0      magrittr_2.0.5      mo
#> [6] farver_2.1.2       vctrs_0.7.3        memoise_2.0.1       askpass_1.2.1       ba
#> [11] tinytex_0.59       htmltools_0.5.9    contentanalysis_1.0.0 curl_7.1.0          br
#> [16] janeaustenr_1.0.0  cellranger_1.1.0   htmlwidgets_1.6.4   tokenizers_0.3.0    pl
#> [21] httr2_1.2.2        cachem_1.1.0       dimensionsR_0.0.3   mime_0.13           li
#> [26] pkgconfig_2.0.3    R6_2.6.1          fastmap_1.2.0       shiny_1.13.0        di
#> [31] patchwork_1.3.2    shinycssloaders_1.1.0 rprojroot_2.1.1     RSpectra_0.16-2     Sn
#> [36] labeling_0.4.3     urltools_1.7.3.1   timechange_0.4.0    mgcv_1.9-1          po
#> [41] httr_1.4.8         compiler_4.4.1     here_1.0.2          bit64_4.8.0         wi
#> [46] S7_0.2.2           backports_1.5.1    viridis_0.6.5       ggforce_0.5.0       MA
#> [51] rappdirs_0.3.4     bibliometrixData_0.3.0 tools_4.4.1         otel_0.2.0          st
#> [56] zip_2.3.3          httpuv_1.6.17      rentrez_1.2.4       nlme_3.1-164        pr
#> [61] grid_4.4.1         stringdist_0.9.17  reshape2_1.4.5      generics_0.1.4      gt
#> [66] tzdb_0.5.0         rscopus_0.9.0      ca_0.71.1           data.table_1.18.4   hma
#> [71] xml2_1.5.2         utf8_1.2.6         ggrepel_0.9.8       pillar_1.11.1       nsy
#> [76] vroom_1.7.1        later_1.4.8        splines_4.4.1       tweenr_2.0.3        la
#> [81] FNN_1.1.4.1        bit_4.6.0          tidyselect_1.2.1    tm_0.7-18           mi
#> [86] knitr_1.51         gridExtra_2.3      NLP_0.3-2           stats4_4.4.1        cru
#> [91] xfun_0.57          graphlayouts_1.2.3 matrixStats_1.5.0   humaniformat_0.6.0  st
#> [96] lazyeval_0.2.3     qpdf_1.4.1         yaml_2.3.12         evaluate_1.0.5      co
#> [101] httpcode_0.3.0     cli_3.6.6          xtable_1.8-8        dichromat_2.0-0.1   Rc
#> [106] readxl_1.4.5       triebeard_0.4.1    XML_3.99-0.23       parallel_4.4.1     ass
#> [111] pubmedR_1.0.2      slam_0.1-55        viridisLite_0.4.3   scales_1.4.0        cra
#> [116] openxlsx_4.2.8.1   rlang_1.2.0        fastmatch_1.1-8

```